

Jishnu Prasad

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EDUCATION

National Institute of Technology Karnataka (NITK Surathkal)

Surathkal, India

B.Tech in Electrical and Electronics Engineering CGPA: 7.18 / 10

Sep. 2024 – Present

EXPERIENCE

SPIRE (Signal Processing, Interpretation and REpresentation) Lab **Indian Institute of Science (IISC)**

Research Intern | Multimodal Retrieval Systems & LLM Evaluation

May 2026 – July 2026

- Built a **multilingual (English & Kannada) multimodal RAG system** for SBI banking forms and documentation, integrating **OCR and web scraping**, alongside a **multi-agent retrieval framework** that dynamically routed queries across PDF, Markdown, Excel, and HTML sources for accurate, context-aware question answering.
- Implemented a **Knowledge-Augmented Generation (KAG)** architecture to overcome limitations of traditional RAG on complex banking queries, enhancing context recall and faithfulness.
- Fine-tuned a Qwen3-27B-Instruct model** on a custom evaluation dataset to act as an answer-quality critic, scoring responses on **Accuracy, Answer Relevancy, and Faithfulness** to enable a **DeepThink feedback loop**

HALE (Healthcare Analytics and Language Engineering) Lab

IT Dept. NITK

Research Intern | Kubernetes, Redis, RAG, Ansible, Raspberry PI

Jan 2026 – Present

- Engineered and operated a multi-node **Raspberry Pi k3s cluster** with **MetalLB and NGINX Ingress**, enabling load-balanced, TLS-secured service delivery.
- Built a full observability stack using **Prometheus, Grafana, Loki, and Promtail** for centralized monitoring and log aggregation.
- Deployed distributed AWS S3 compatible object storage with **MinIO**, Flask/React applications, and **JupyterHub**, addressing multi-node storage, image distribution, and service discovery challenges.
- Ansible**-based provisioning and orchestration for reproducible edge inference workloads on a Raspberry Pi 5 Kubernetes cluster.

Urban Records

Remote

Freelance Full-Stack Engineer | Django, ReactJs, Docker, PostgreSQL, Dokploy

Sep. 2025 – Nov. 2025

- Built and deployed a containerized **full-stack** document management platform using **Django REST Framework and React** serving **500+ users** and reducing document processing time from days to hours.

SENA (Shiksha Evam Naitik Abhiyaan) NGO

Bengaluru

Backend Engineering Intern & Convenor NITK Chapter | Django, Bootstrap, Docker

May 2025 – Jul. 2025

- Developed and integrated a secure donations pipeline using the Cashfree Python SDK and built form-driven publishing workflows, enabling reliable payment processing, transaction tracking, and self-service content publishing for non-technical contributors.

PROJECTS

Memora — Redis compatible In-Memory Database | *Go, Python, TCP, Concurrent Systems*

<https://github.com/Jishnu-Prasad888/Memora>

- Built a Redis-compatible in-memory key-value database in Go with **RESP protocol** support, concurrent TCP client handling, custom **hash-table storage**, **TTL expiration**, and **RDB-style snapshot persistence**.
- Developed CLI and Python GUI clients for administration and monitoring, achieving **226K+ ops/sec** over **1M benchmarked operations** with **39µs P50** and **2.0ms P99** latency under concurrent workloads.

ATLAS — Distributed Monitoring and Observability Platform | *ReactJS, Django, NATS JetStream, AI, Redis, Cloudflare*

<https://github.com/Jishnu-Prasad888/ATLAS>

- Built a distributed **Linux fleet management and observability platform** for managing servers without direct SSH access, using a centralized **ATLAS** control plane and lightweight **Beacon** agents.
- Engineered secure **NATS JetStream** communication with durable delivery, offline buffering, message replay, dead-letter queues, and **mTLS**, enabling resilient remote fleet operations.
- Developed real-time observability for **CPU, RAM, disk, network, processes, Docker, and Kubernetes**, with historical metrics, live telemetry streaming, diagnostics, and centralized log analysis.
- Implemented multi-tenant **RBAC**, immutable audit logging, process/container management, network diagnostics, and **Multi-Agent AI-assisted** log, telemetry, anomaly, and root-cause analysis; deployed with **Docker, Nginx, Cloudflare Tunnel**, and **Vercel**.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript C++, Go

Frameworks & Libraries: Django, FastAPI, React, Next.js, LangChain, PyTorch, OpenCV

Tools & Platforms: Linux, Docker, Kubernetes, PostgreSQL, Redis, Git, MCP, RAG, AWS (basics)

LEADERSHIP & EXTRACURRICULARS

Top 10 / 1,922 Teams — Caterpillar Autonomy Challenge, IIT Madras

Team Member, NITK CATERBOTS

Amateur Astronomy Club (AAC) — Chief Technical Coordinator

Open Source Contributions to Homelab-Monitor & Tabularis; GSSoC'25,'26

Smart India Hackthon(NITK Internal Round) Top 1 in problem statement